

DPP - 01

CS & IT

Algorithms

Dynamic Programming (DP)

Q1 What is the time complexity of dynamic programming for matrix chain multiplication problem ?

- (A) $O(n^2)$ (B) $O(n^3)$
(C) $O(n \log n)$ (D) None of these

Q2 Consider the matrices x, y and z with dimension 10×20 , 20×30 and 30×40 respectively. Then what is the minimum number of multiplications required to multiply the matrices ? _____

Q3 What is the length of the LCS for the pair of strings given below

P = ATGACTATAA

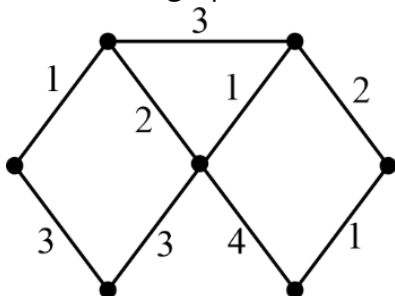
Q = GACTAATA

- (A) 5 (B) 6
(C) 7 (D) 8

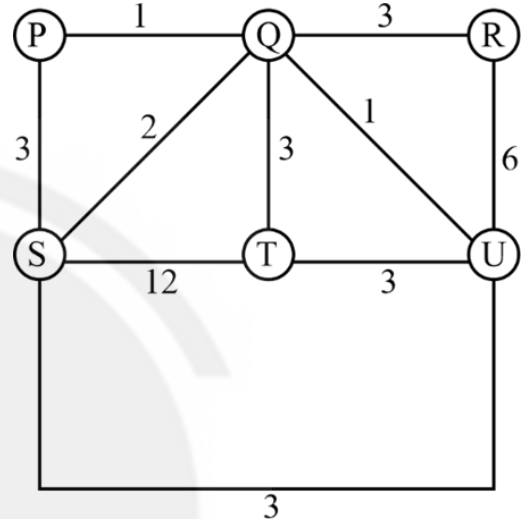
Q4 Consider a connected weighted graph $G = (V, E)$, where $|V| = n$, $|E| = m$, if all the edges have distinct positive integer weights, then the maximum number of minimum weight spanning trees in the graph is ?

- (A) n (B) m
(C) 1 (D) n^{n-2}

Q5 What is the weight of the minimum spanning tree for the graph shown below?



Q6 How many minimum spanning tree does this graph have?



- (A) 2 (B) 3
(C) 4 (D) 5

Q7 Consider the following problem with knapsack capacity of 8

Items	Profits	Weights
I_1	13	1
I_2	8	5
I_3	7	3
I_4	3	4

Which of the following item is not selected in the optimal solution of 0/1, knapsack problem?

- (A) I_1 only (B) I_2 only
(C) I_3 only (D) I_4 only

Q8 Consider the following statements:

S_1 : for every weighted graph and any two vertices p and q, Bellman ford algorithm starting at p will always return a shortest path to q.



S_2 : Dijkstra greedy algorithm for Single source shortest path can be used to solve the all pairs shortest path problem.
Which of the statement is correct?

- (A) only S_1
- (B) only S_2
- (C) Both S_1 and S_2 are true
- (D) neither S_1 nor S_2 is true



Answer Key

Q1 B
Q2 18000
Q3 C
Q4 C

Q5 10
Q6 A
Q7 B
Q8 B



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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Time complexity of the matrix chain multiplication problem in dynamic programming approach is $O(n^3)$. Where n is the number of matrices. Because it contains nested loop iterating over the matrix dimension to fill in the optimal costs.

Q2 Text Solution:

Given matrix dimension

$x : 10 \times 20$

$y : 20 \times 30$

$z : 30 \times 40$

Optimal parentification is $((xy)z)$

Minimum number of multiplications

$= 10 \times 20 \times 30 + 10 \times 30 \times 40 = 18000$

Q3 Text Solution:

$P = A T G A C T A T A A$

$Q = G A C T A A T A$

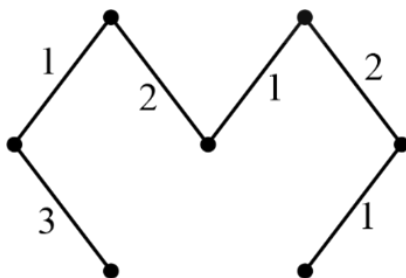
Subsequence 1 = $G A C T A T A$

Subsequence 2 = $G A C T A A A$

Q4 Text Solution:

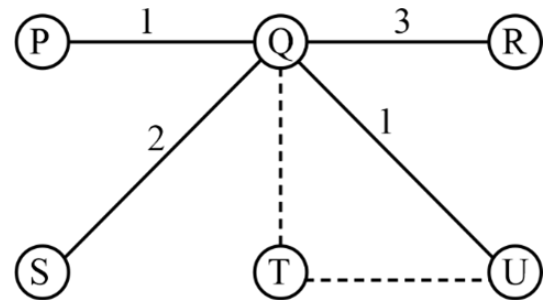
The maximum number of minimum weight spanning tree in a connected weighted graph $G(V, E)$ with $|V| = n$ and $|E| = m$, where all edges have distinct positive integer weight is 1.

Q5 Text Solution:



$$1 + 2 + 1 + 2 + 3 + 1 = 10$$

Q6 Text Solution:



There are 2 dotted lines are the only choices that it has,

Hence are only 2 MST possible here.

Q7 Text Solution:

$w = 8(\text{capacity})$

Feasible solution

(i) $\{I_1, I_3, I_4\}$

Profit = $13 + 7 + 3 = 23$

(ii) $\{I_2, I_3\}$

Profit = $8 + 7 = 15$

Optimal solution = $\{I_1, I_3, I_4\}$

With the capacity of 8 and maximum profit produced is 23.

I_2 is not selected in the solution.

(b) is correct option.

Q8 Text Solution:

Bellman ford algorithm may not return a shortest path from p to q if there exist -ve edge weight cycle.

S_1 is false.

Dijkstra greedy algorithm for SSSP can be used to solve the all pairs shortest path problem. **TRUE**



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